Sim2Real Transfer Learning for Nanobody Thermal Stability Prediction

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1 Background/Objectives

VHH antibodies (nanobodies) are small and easy to engineer, making them promising therapeutic candidates[1]. However, their thermal stability is sequence-dependent, and experimental measurement of the melting temperature (T_m) is costly. While molecular dynamics (MD) simulations can compute stability-related quantities such as $\Delta\Delta G$, these calculations are computationally demanding and require significant HPC resources.

Sim2Real transfer learning[4], which leverages large-scale simulation data to enhance predictions on limited experimental data, has proven effective in materials science. We apply this approach to nanobody stability prediction. Since experimental T_m and simulation-derived $\Delta\Delta G$ values cannot be directly combined, we propose to use multitask learning with shared representations to bridge the gap between simulation and real-world data.

2 Methods

Protein sequences were encoded using the protein language model ESM-2[3], and the resulting representation vectors were used as inputs to the predictive model. As the real experimental dataset, we employed NbThermo[5] (567 samples), which consists of melting temperature (T_m) measurements for VHH antibodies. As simulation data, we incorporated $\Delta\Delta G$ values (846 samples) computed by free energy perturbation (FEP) based on all-atom molecular dynamics simulations, and those (846 samples) computed by Rosetta[2], respectively.

The model was trained in a multitask learning framework to enable Sim2Real transfer. A shared MLP backbone was used to learn a common latent representation across tasks, with task-specific heads for real T_m prediction and simulation-derived $\Delta\Delta G$ prediction.

The multitask loss function was defined as:

$$\mathcal{L} = \frac{1}{|S_{T_m}|} \sum_{i \in S_{T_m}} \left(\hat{y}_i^{(T_m)} - y_i\right)^2 + \frac{1}{|S_{\Delta\Delta \mathrm{G}}|} \sum_{i \in S_{\Delta\Delta \mathrm{G}}} \left(\hat{y}_i^{(\Delta\Delta \mathrm{G})} - y_i\right)^2 .$$

During training, 56 samples were randomly drawn from the 567 T_m data points to form the multitask training set, emulating a low-data regime. This procedure was repeated 100 times with different random splits, and the average performance over the 100 runs was reported.

3 Results

Multitask learning consistently outperformed the single-task baseline (a model trained only on real T_m data). The dependence of the prediction error on the number of FEP simulation samples n can be approximated by the following scaling law:

$$R(n, 56) = 0.045660\, n^{-0.419244} + 7.256781.$$

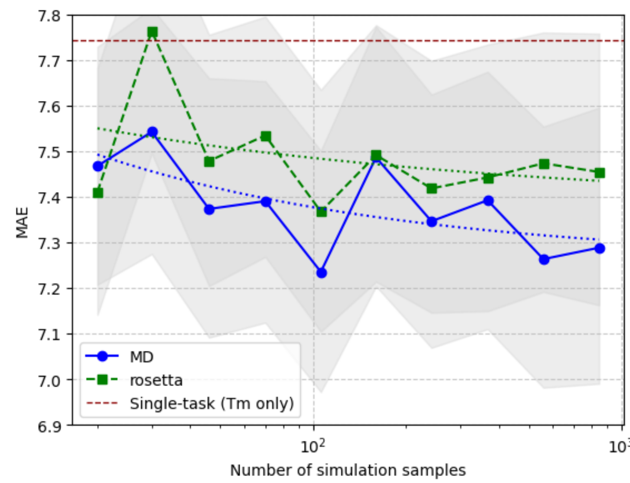

Figure 1: Comparison between single-task (T_m only, real data) and multitask learning (real T_m + simulation $\Delta\Delta G$).

Based on the concept of marginal rate of substitution[4], we estimated that approximately 343 FEP simulation samples provide an equivalent improvement to a single experimental T_m sample.

4 Conclusions

We demonstrated Sim2Real transfer learning for nanobody thermal stability prediction by integrating simulation-derived $\Delta\Delta G$ with experimental T_m through multitask learning. The estimated marginal rate of substitution (343:1) provides a quantitative guideline for leveraging simulations to compensate for scarce experimental data, and can inform efficient allocation of HPC resources in biomolecular simulation campaigns.

References

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